



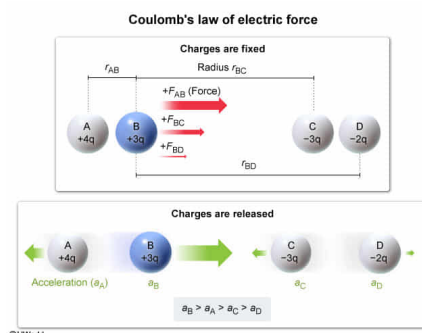
Four charged particles of equal mass are initially fixed in place along a straight line, as shown above. If all the charged particles are released at the same time, which particle would initially accelerate most rapidly?

- ☐ A. Particle A [35%]
- ✓ ☒ B. Particle B [41%]
- ☐ C. Particle C [12%]
- ☐ D. Particle D [10%]

Correct

41% answered correctly

Explanation:



The sign (positive or negative) and quantity of electric charge influences the way [charged particles interact](#). For example, charges with the same sign exert repulsive forces against each other whereas charges with opposite signs are attracted to one another.

Coulomb's law states that the electric force (F) between two charges is directly proportional to the charge of each particle (q_1, q_2) but inversely proportional to the square of the distance (r) separating the charges. This relationship is expressed as an equation via Coulomb's constant ($k = 8.99 \times 10^9 \text{ N}\cdot\text{m}^2\cdot\text{C}^{-2}$):

$$F = k \frac{q_1 q_2}{r^2}$$

In this question, each of the four charges fixed along the line experiences an electric force from the other three nearby charges. Depending on the sign of each charge, the multiple electric forces within this setup are either attractive or repulsive.

Specifically, particle B experiences three forces: a repulsive electric force from particle A (F_{AB}) and attractive electric forces from both particle C (F_{BC}) and particle D (F_{BD}). Each of the forces acting on particle B is oriented along the same line toward particles C and D (ie, to the right). Furthermore, F_{AB} has the largest magnitude in this setup because the charges of particles A and B (+4q and +3q, respectively) are relatively large and the distance between particles A and B (r_{AB}) is relatively small.

Consequently, the sum of electric forces acting on particle B is the largest in magnitude such that particle B accelerates most rapidly once the charges are released.

(Choice A) Particle A will not accelerate to the left as rapidly as particle B accelerates to the right because particle A is also attracted to particles C and D, which are located to the right of particle A.

(Choices C and D) Neither particle C nor D will accelerate as rapidly as particle B because the electric charges of particles C and D are smaller in magnitude than the charges of particles A and B.

Educational objective:

The magnitude and direction of the attractive or repulsive forces exerted between electric charges are directly proportional to the charge of each particle but inversely proportional to the square of the distance separating the charges.

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